

in a vehicle is not burnt properly due to bad air to fuel mixture. The lack of transistor controlled ignition (TCI) means that you cannot use twin spark plugs. Overall, TCI alone could generate a 12 per cent increase in efficiency. With the move to Bharat Stage VI emission standards, the Indian government has increasingly started focusing on better pollution control. Therefore the semiconductors that ROHM mentioned earlier will start playing a major role in controlling the emissions, which will in turn control the pollution output of these vehicles. The emission control for Euro 6 will require a big improvement in efficiency, and that is where the ECU will play a major role.

ROHM also expects to see a drastic change in capacitor discharge ignition (CDI) transforming into transistor controlled ignition (TCI), and they believe that the new solutions will be innovative and cost effective for Indian customers. Their automotive segment comprises mainly of components targeting TCI and LED clusters.

Growth of Power Electronics: High-end two wheelers are also an area that will be a driving factor in the market due to their use of electronics and battery power. ROHM found that they could win in the India market using their analogue power components. "We have support systems already setup, and we even support IITs on the 48-volt initiative. We believe that this is one of the ways in which we can contribute to society, and power electronics will be a big part in all of this," adds <spokesperson>.

Other plans

As ROHM grows, they are looking at expanding operations and human

THE PRODUCT INNOVATION SEMINAR

Keeping in line with the company's aim to drive innovation in product design without sacrificing on quality, ROHM Semiconductor conducted their Product Innovation Seminar in Chennai on the 15th of March, 2016.

Seminars were delivered by some of the key speakers from ROHM. "Electronics Innovation for Automotive," was the theme focused on by Mr.Yoshiaki Suenaga, General Manager, Automotive Strategy Division, ROHM Co. Ltd., Japan. Here he spoke about the future of semiconductor development, Advanced Driver Assistance Systems (ADAS), Human Machine Interfaces (HMI), and the electrification of powertrains. The second theme was on ROHM Semiconductor's "Vision for India," delivered by Mr.Vijay Bolloju, Design Centre Manager, ROHM Semiconductor India Pvt. Ltd.

man resources for their applications engineering area as well as customer support area. Distribution through online will also be one of the key areas that they aim to drive.

Enjoying the benefits of being a vertically integrated company

Being a vertically integrated manufacturing company, ROHM is able to innovate and make their own materials and packaging technology. Innovation in new semiconductor processes and packaging technologies is a key part of what ROHM drives to ensure that they are able to push the barriers on lower power consumption and compact foot print that the market requires.

For future, materials like SiC and GaN, ROHM was able to improve the process and therefore build far better products. For example, they were able to use the Super-junction technology in MOSFETs to improve the efficiency as compared to the previous planar MOSFET technology.

In-house packaging at ROHM has been tweaked from the original packaging to enhance the downsizing and improve package power. Material innovation is a key element driven by

ROHM, and SiC is a good example of this for the motor industry.

Additionally, ROHM has experience from growing silicon, to fabrication and then with packaging. This lets them control the quality and material through out the process.

ROHM's vision for 2020

ROHM says that they want to contribute in building a comfortable society in India, enabled by their products. Automotive, home appliances, and communication technologies will be the core to achieve this. They also aim to improve RF-Sub GHz communication system's interoperability through the Wi-SUN (Wireless Smart Utility Network) alliance, and leverage IoT industry standards to make headway in Home Energy Management Systems (HEMS) and Building Energy Management Systems (BEMS).

The company envisions an increase in communications technologies and MEMS sensors as the Internet of Things moves forward. Moreover, the research and development for smartphones seems to be shaping up a lot in India, and ROHM believes that we will soon see not just Made in India phones, but Designed in India phones as well.